Recurrent networks

Chicago transit ridership

LECTURE 16 | GÉRON CH. 13

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

Where we are

Lecture 15 set the ground rules on this same series, and a number to beat:

- Evaluation is **forward in time**. A shuffled split puts a day's own neighbours in training, and backtesting from a rolling origin is what a forecast is scored by
- Copying the same weekday last week is genuinely hard to beat, because the autocorrelation at lag seven is large: its error is the seasonal difference, and that difference is small
- A linear model on the last 56 days contains that baseline — one weight of one on lag seven — and beats it, modestly

> **WHAT A LINEAR MODEL ON LAGS CANNOT DO**
>
> It has a **fixed window**. Anything outside the last 56 days is invisible to it, and every lag gets one weight regardless of context. A recurrent network keeps a state instead, and that state is not bounded by a window.

Today

1. **The recurrent cell** — one equation, four shapes of model, backpropagation through time, and a first network trained and measured
2. **The method** — the loss, the windows, the scale, three ways to forecast ahead; then a deeper network, more series, and fourteen days at once
3. **Gates** — LSTM and GRU, the ladder of every model on the same days, and what March 2020 did to a model fitted on the years before

No new derivation today. Lecture 11's variance argument is the one that applies, unrolled through time instead of down layers — which is exactly why the gates exist. the architectures, and what each gate is for.

EXAMINABLE

Open Lecture 16 in Colab

The recurrent cell, written out

$$\mathbf{h}_t = \phi\left(\mathbf{W}_x^\mathsf{T}\mathbf{x}_t + \mathbf{W}_h^\mathsf{T}\mathbf{h}_{t-1} + \mathbf{b}\right)$$

ONE STEP OF A RECURRENT LAYER

- $\mathbf{x}_t$ — the input at step t; $\mathbf{h}_t$ — the state after it
- $\mathbf{W}_x$ and $\mathbf{W}_h$ do not depend on t: **the same weights at every step**
- Compare Lecture 9: this is $\phi(\mathbf{XW} + \mathbf{b})$ with one extra term, and that term is the whole idea

The state is the only route by which anything before step t reaches step t. Its width is therefore a budget: whatever the network needs to remember has to fit in $\mathbf{h}$.

Four shapes of recurrent model

Shape	Input	Output	Example
sequence to vector	a sequence	one value	**✓ today's model**
vector to sequence	one value	a sequence	a caption from an image
sequence to sequence	a sequence	a sequence	a forecast at every step
encoder–decoder	a sequence	a sequence	translation

We build the first and the third today. The fourth is Lecture 18.

Training it: backpropagation through time

Unroll the network over the window and it becomes a deep feed-forward network whose layers *share their weights*. Then differentiate it exactly as Lecture 10 taught — nothing new is needed.

> **WHICH IS ALSO THE PROBLEM**
>
> Unrolling 56 steps gives a 56-layer network, and Lecture 11 said what happens to a gradient descending 56 layers: it is multiplied by the same factor 56 times. Here the factor is the *same* matrix every step, because the weights are shared — so the gradient is governed by a power of one matrix, and powers of a matrix either vanish or explode unless its spectral radius sits almost exactly at one.

Initialisation cannot fix this the way it did in Lecture 11, because there the layers were independent. Something has to change in the cell itself.

The model, in eleven lines

```
import torch.nn as nn

class SimpleRnn(nn.Module):
    def __init__(self, input_size=1, hidden_size=32, output_size=1):
        super().__init__()
        self.rnn = nn.RNN(input_size, hidden_size, batch_first=True)
        self.head = nn.Linear(hidden_size, output_size)

    def forward(self, X):                       # X: [batch, time, series]
        outputs, last_state = self.rnn(X)
        return self.head(outputs[:, -1])        # only the final step
```

`batch_first=True` because our batches are [batch, time, series]. Without it the module silently reads the first dimension as time.

Training it

```
model = SimpleRnn()
loss_fn = nn.HuberLoss(delta=0.05)
opt = torch.optim.SGD(model.parameters(), lr=0.05, momentum=0.9)
loader = DataLoader(train_set, batch_size=32, shuffle=True)

for epoch in range(200):
    model.train()
    for Xb, yb in loader:
        opt.zero_grad()
        loss_fn(model(Xb), yb).backward()
        opt.step()
```

The same five lines as Lecture 10, unchanged. `zero_grad` is still not optional and still fails silently.

Why the Huber loss

$$L_\delta(e) = \begin{cases} \frac{1}{2}e^2 & |e| \le \delta \\ \delta\left(|e| - \frac{1}{2}\delta\right) & |e| > \delta \end{cases}$$

Quadratic near zero, linear far away. We are optimising a proxy for MAE, and minimising the absolute error directly gives a gradient that never gets smaller as you approach the optimum.

The metric and the training loss are allowed to differ: the metric is what the operator reads, the loss is what the optimiser can descend. Here is the case.

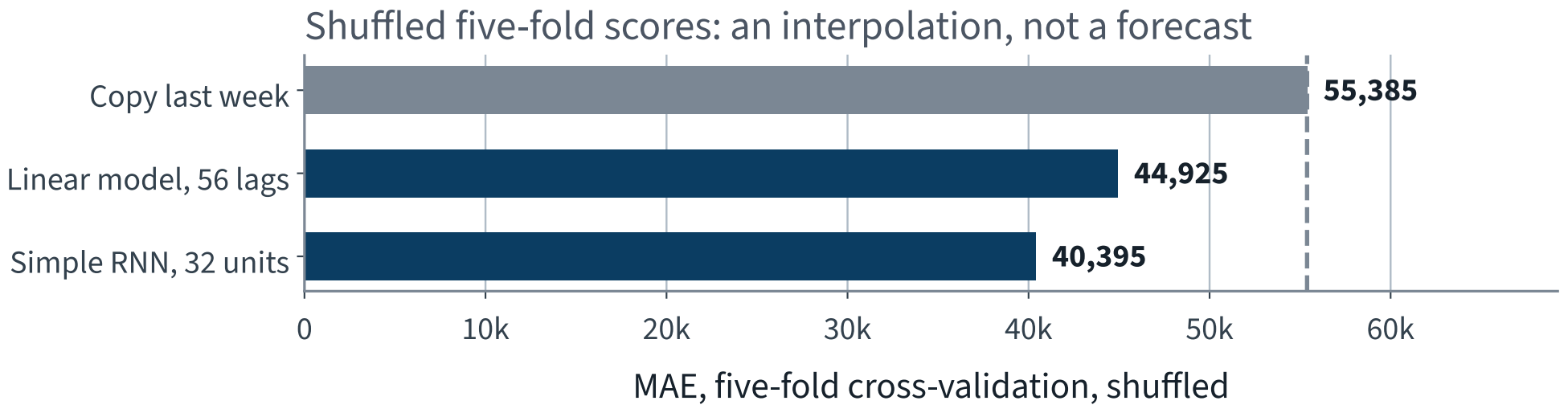
The recurrent network, measured the same way

40,395

Five folds, shuffled, seed 42 — identical protocol to the linear model.

Against **55,385** for copying last week on the same folds and **44,925** for the linear model (Lecture 15's, averaged over twenty shuffles of five folds). That is **27.1%** better than the baseline, and 10% better than the linear model.

Two models, under the same shuffled protocol



Two models, both comfortably past the baseline, under the same shuffled protocol.

✗ These folds are shuffled, so none of these numbers is a forecast. Every forward number in this lecture is scored on the same 151 days.

Loss for a forecast

Squared error punishes a large miss quadratically, which on a series with occasional shocks means a handful of days dominate the gradient.

Loss	Behaviour
MSE	smooth everywhere; dominated by outliers
MAE	robust; gradient is constant, so it converges slowly near the optimum
Huber	quadratic near zero, linear beyond — both properties, one knob

The choice is the same argument as choosing the metric in Lecture 15, and it has the same answer: it depends on what an error costs the operator, and that is a question for them and not for us.

Windows overlap, and that matters

Consecutive training windows share almost all their content: a 56-day window starting on day t and one starting on day $t + 1$ differ in two values out of fifty-six.

TWO CONSEQUENCES

Shuffle them. Otherwise a mini-batch is near-identical examples, and averaging near-copies cancels almost none of one example's noise: the step is noisier, and training stalls.

Do not let that shuffling cross the split. Windows are shuffled *within* the training period only; the evaluation period is still strictly later in time. Shuffling and leakage are different operations and it is easy to conflate them.

Normalising a series

- Scale by a constant computed on the **training period only** — the same rule as every other lecture, and easier to break here because the series looks like one object
- Divide rather than standardise if the series is positive and multiplicative: ridership is counts, and a ratio is more natural than a difference
- If you differenced, remember to *un*-difference the prediction before reporting it. The metric is in the units the operator uses

✗ A model evaluated in differenced units can look excellent and mean nothing: predicting a change of zero every day is nearly right, and it is the naive forecast wearing a disguise.

> **FAILURE CONDITION — A MODEL SCORED IN DIFFERENCED UNITS**
>
> A model evaluated on differences can look excellent and mean nothing: predicting no change at all scores well in those units. The units the metric was computed in decide what it is able to praise, and the report is usually silent about which they were.

What to log while it trains

- Training and validation loss per epoch — the gap is Lecture 5's
- The gradient norm — Lecture 11's probe, along time here rather than depth. A plain RNN on a long window shows it decaying
- The forecast itself, plotted against truth, on a held-out stretch. A number cannot tell you the model has learned to output last week’s value with a one-day lag; a picture can

> **THE FAILURE A PLOT CATCHES AND A METRIC DOES NOT**
>
> A model that simply repeats the previous value scores well on almost any error metric and is worthless. On a plot it is unmistakable: the prediction is the truth, shifted right by one step.

Forecasting k steps ahead: three strategies

Strategy	How	Weakness
Recursive	predict one step, feed it back, repeat	errors compound; the model never trained on its own output
Direct	a separate model per horizon	k models, and none of them shares what the others learned
Sequence-to-sequence	one model, k outputs at once	more to fit, and the outputs are not independent

All three are defensible and all three must be evaluated the same way: **forward in time, at the horizon you actually need**. A model that is excellent one step ahead and useless seven steps ahead is not a good model if the operator plans a week in advance.

Teacher forcing, and what it costs

Training a sequence-to-sequence model, you can feed the decoder the *true* previous token or its *own* previous prediction.

	Trains	At prediction time
True token (teacher forcing)	**fast and stable ✓**	**✗ the model has never seen its own mistakes**
Own prediction	**✗ slow, and early training is noise**	**matches deployment ✓**

EXPOSURE BIAS

A model trained only on true prefixes meets a distribution at prediction time that it never saw in training: its own errors, compounding. The forecast drifts, and the further ahead you go the worse it gets — which is precisely the recursive strategy's weakness on the previous slide.

A model trained only on true prefixes

FAILURE CONDITION

Trained on true history and evaluated on its own output, the model meets a distribution at prediction time it never saw in training. Errors compound step by step, and the training loss — computed on true prefixes throughout — never registers it.

Improvement 1 · A deeper network

```
self.rnn = nn.RNN(input_size, hidden_size, num_layers=3,
                  batch_first=True)
```

One argument. Three recurrent layers stacked, the output of each feeding the next.

Model	Forecast MAE	Train MAE	Recipe
Simple RNN, one layer	**✗ 58,029**	37,388	adam
Deep RNN, three layers	**45,912 ✓**	31,359	adam

On this run, depth won — on test *and* on train. **Do not write that down yet.**

The same comparison, run twice

Each recipe is scored on July–December 2018, inside the training period, and the lower score wins. For the simple RNN:

	sgd	adam	Chosen	Simple RNN	Deep RNN
This run	41,878	40,910	adam	✗ 58,029	45,912 ✓
The deck's original run	41,878	42,879	sgd	49,073 ✓	✗ 53,491

- The choice is a near-tie — 968 riders here, about 1,000 the other way before, 2.3% either way — and it flipped
- The sgd score reproduced to the rider; the adam fits of both univariate RNNs did not, so the deep RNN moved too, on the same recipe. The linear and multivariate rows reproduced exactly
- The original run read "depth is worse on train too (40,630 against 35,972), so a harder optimisation". This run says the opposite, with the same code

✓ A ranking that rests on one near-tied selection decision is not a finding. Report it as unresolved, and settle it only with several seeds and a gap larger than their spread.

Improvement 2 · Give it more series

We have been forecasting rail from rail alone, while a bus series sits unused beside it — and a day-type column we established is known in advance.

```
mulvar = df[["rail", "bus"]] / 1e6
mulvar["next_day_type"] = df["day_type"].shift(-1)   # known in advance
mulvar = pd.get_dummies(mulvar, dtype=float)         # 5 columns

assert list(mulvar.columns) == ["rail", "bus", "next_day_type_A",
                                "next_day_type_U", "next_day_type_W"]
```

The architecture changes by one number: `input_size=5`.

Wait — that is a `shift(-1)`

The previous lecture measured a random split that let a model train on the rows surrounding what it was scored on. This one reads the future *deliberately*.

> **THE TEST THAT SEPARATES THEM**
>
> Is the value **available at prediction time**? The type of the day we are forecasting is on the public calendar and has been for a year. Its ridership is not.

✗ This is the one question about leakage that no static check can answer for you. It is not a property of the code; it is a property of the world.

The result

Model	Forecast MAE	vs copying last week
Copy last week	64,815	—
Simple RNN, rail only	58,029	10% better
Simple RNN, five inputs	**40,046 ✓**	38% better

✓ Thirty-one per cent off this run's rail-only error for one line of feature engineering and one changed constructor argument.

Against the original run's rail-only row it was eighteen per cent — the rail-only row is the unstable one. The five-input row reproduced to the rider, and it beats both.

Most of it is the day type. The network no longer has to infer that a holiday is coming from the shape of the previous eight weeks — which is impossible — because we told it.

Improvement 3 · Forecasting further than one day

The operations team also wants a fortnight. Two ways:

Feed the forecast back

Predict day $t+1$, append it to the window as though it had happened, predict $t+2$. Errors accumulate.

Predict all fourteen at once

Change the output layer to fourteen units and the target to a vector. One model, no accumulation.

```
target = self.series[end : end + 14, 0]     # 14 values, not 1
model = GruModel(input_size=5, hidden_size=32, output_size=14)
```

And a third way — predict fourteen at *every* step

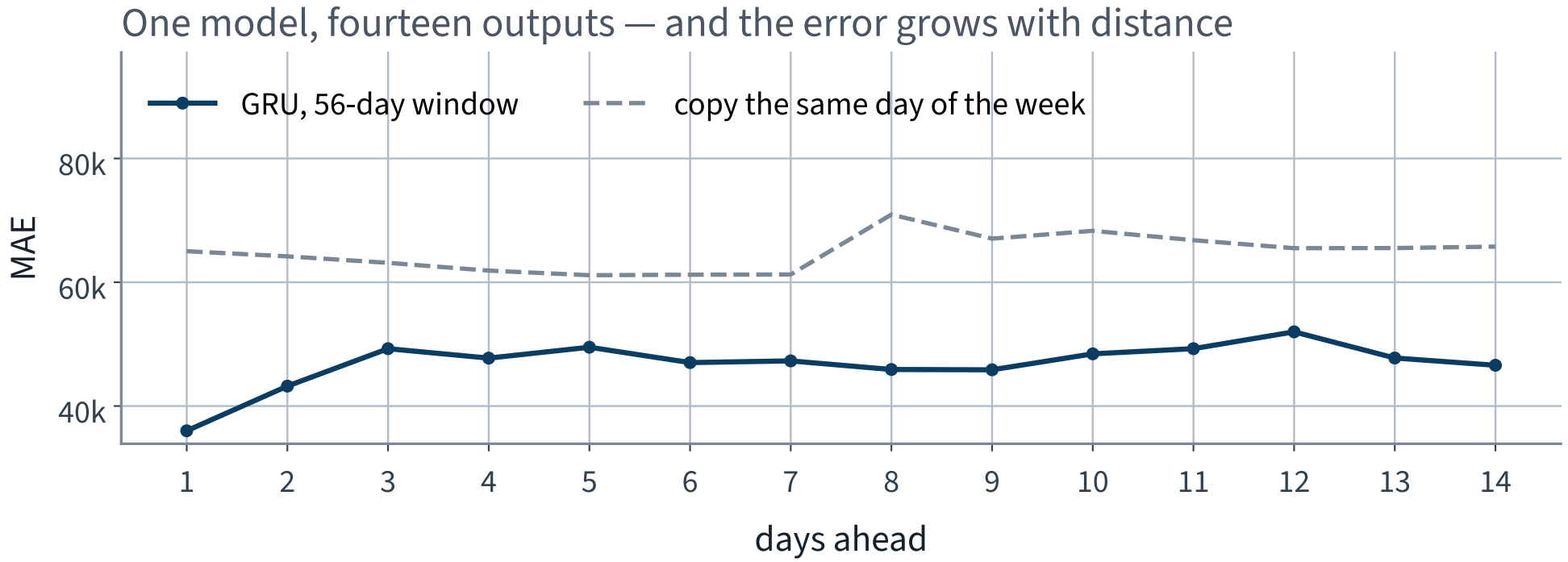
At step 0 the model outputs the forecasts for steps 1 to 14; at step 1, for steps 2 to 15; and so on. The target is a sequence of windows rather than a single vector.

```
def forward(self, X):
    outputs, _state = self.rnn(X)
    return self.head(outputs)        # every step, not outputs[:, -1]
```

> **IS THAT CHEATING?**
>
> No. A recurrent cell at step t has seen steps 1 to t and nothing more — it is **causal**. The loss simply gains a term at every step, so many more gradients flow and they travel a shorter distance.

After training, only the last step's output is used.

How far ahead can it see?



The error grows with distance, from **35,997** at one day to **46,586** at fourteen — and stays below the no-model baseline throughout.

Why a gate helps, in one line

A plain cell *multiplies* the state at every step:

$$\frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_{t-1}} = \operatorname{diag}(\phi')\,\mathbf{W}_h^{\mathsf{T}}$$

so over T steps the gradient carries T copies of the same matrix, and Lecture 11's argument applies with the extra force that it is *one* matrix rather than many independent ones.

WHAT A GATED CELL DOES INSTEAD

An LSTM keeps a separate cell state $\mathbf{c}_t$ updated **additively**: $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$. When the forget gate $\mathbf{f}_t$ is near one, $\partial \mathbf{c}_t / \partial \mathbf{c}_{t-1} \approx \mathbf{1}$ and the gradient passes through almost unchanged. That is the entire trick: **a path along which the gradient is added to rather than multiplied.**

LSTM, gate by gate

Gate	Equation	What it decides
forget $\mathbf{f}_t$	$\sigma(\cdot)$	how much of the old cell state to keep
input $\mathbf{i}_t$	$\sigma(\cdot)$	how much of the candidate to write
candidate $\tilde{\mathbf{c}}_t$	$\tanh(\cdot)$	what to write, if anything
output $\mathbf{o}_t$	$\sigma(\cdot)$	how much of the cell state to expose as $\mathbf{h}_t$

Every gate is a sigmoid of the same two inputs $(\mathbf{x}_t, \mathbf{h}_{t-1})$ with its own weights, so a sigmoid in $[0, 1]$ is a soft switch and the network *learns when to open it*. Four gates means roughly four times the parameters of a plain cell.

State what each gate does and why the cell-state update is additive.

EXAMINABLE

GRU: the same idea, two gates

- **Update gate** $\mathbf{z}_t$ — merges LSTM's forget and input into one: what is kept is what is not written
- **Reset gate** $\mathbf{r}_t$ — how much of the previous state to use when forming the candidate
- No separate cell state: $\mathbf{h}_t$ does both jobs

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$$

Still additive, still a near-identity path when $\mathbf{z}_t$ is small — the property that mattered, kept, with about three quarters of the parameters.

> **WHICH TO USE**
>
> On most problems they are within noise of one another. GRU is smaller and trains faster; LSTM has the longer track record. Measure, on your own protocol, rather than inheriting an opinion.

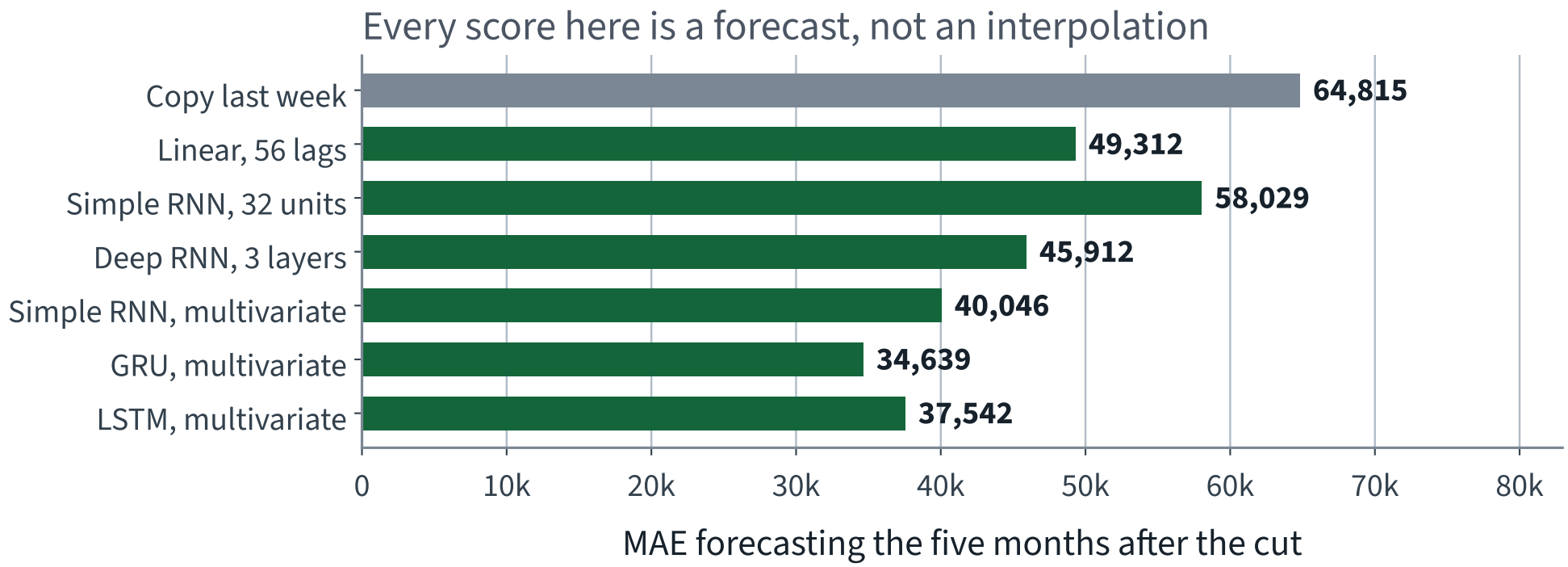
Both gated cells, measured

Model, all five inputs	Forecast MAE	Training MAE
Simple RNN	40,046	27,035
LSTM	37,542	24,540
GRU	**34,639 ✓**	**✗ 20,356**

The GRU wins on the forecast, and the second column says why to be careful: it fits the training rows 25% better than the plain cell but the forecast only 14% better. **That gap is overfitting**, and it is what to attack next, before reaching for a bigger cell.

Both cells are Chapter 13. Neither is a large model: a few thousand parameters.

Every model, measured the honest way



From 64,815 to 34,639 — **47%** better than doing nothing, and every bar is a forecast on the same days.

The whole application, in one table

Forecast	What we reported	What it does
Copy last week	55,385	64,815
Linear, 56 lags	44,925	**✗ 49,312**
Simple RNN	40,395	**✗ 58,029**
GRU, five inputs	—	**34,639 ✓**

The two middle rows are the same models measured twice; the simple RNN's right-hand number is the one that moves between runs. The linear row's left number is Lecture 15's scikit-learn fit; its right number is this notebook's, on the same 151 days as every other right-hand number. The bottom row is the model we would actually deploy, and it was never measured the wrong way.

✓ The final number is better than the one first reported — but we reached it by first admitting that the reported one was fiction.

Is 34,639 good?

It is **47%** better than the system a transit authority could run with no data scientist at all, and its typical miss is around 10.4% of the day it is forecasting — against 16.5% for copying last week *on those same 151 days*. A naive percentage measured over any other set of days is not the comparison to make here. (As a share of a mean day on those same 151 days — 583,030 riders — the miss is 5.9%, which is a third quantity again.)

> **WHAT A DEFENSIBLE REPORT SAYS**
>
> “Forecasting the five months after the training cut, and never seeing them, the model’s mean absolute error is 34,639 boardings a day against 64,815 for copying the same day of the previous week. Both figures are on identical rows. The gain is concentrated on holidays.”

Note what that sentence contains: the protocol, the baseline, the rows, and where the gain comes from. Four things the number alone does not carry.

Red team it: what happens when the world changes?

Fit on everything to the end of 2019. Forecast the four months from March 2020, which nobody in this room needs reminding about.

	MAE
The linear model, trained to the end of 2019	✗ 84,380
Copying last week, on the same 122 days	36,282 ✓
The actual mean level over that period	141,404

✗ The model is 2.3 times worse than doing nothing, and its error is six tenths of the entire level.

> **FAILURE CONDITION — A REGIME THE TRAINING WINDOW NEVER CONTAINED**
>
> Scored on the four months from March 2020, the model is 2.3 times worse than doing nothing and its error is six tenths of the level of the series. Nothing in training could have shown this: the failure is that the future stopped resembling the window the model was fitted on.

To be fair to the model

March 2020 is not a leak and it is not the model's fault. No forecaster could have produced it from eight weeks of history, and copying last week wins only because it adapts within a week of the collapse.

> The lesson is not "the model is bad". It is that an honest protocol *told you* the model had failed, on the day it failed. A shuffled split, with those same days spread across its training folds, could not have.

✓ Which is why a deployed forecaster is monitored against its own naive baseline, continuously. If the margin disappears, something has changed in the world.

Temporal leakage checklist — keep this

- Is the split by **date**, not by row?
- Does every feature exist at the moment the forecast is made? Name the moment
- Were any statistics — means, scalers, encodings — computed over the whole series before the cut?
- Do the training windows overlap the scored windows? Purge them
- Is the baseline computed on *identical rows* to the model?
- Does the reported score come from more than one origin?

Six questions. The third is the oldest leak in the course — a statistic fitted before the split — and the fourth is the one only a series of overlapping windows has.

Reading a recurrent result honestly

Question	Why it decides whether the number means anything
At what horizon?	one step ahead is a different problem from seven
Against which baseline?	seasonal-naive, not the mean — Lecture 15
On which protocol?	rolling origin, and how many folds
With what spread?	one origin is one sample
In which units?	differenced or not; scaled or not

✗ A recurrent network that beats the naive forecast by less than the fold-to-fold spread has not beaten it — and, as the depth comparison showed, a ranking that flips between runs was never a ranking.

A recipe for a forecasting model

1. Plot the series. Seasonality and trend are usually visible, and cost nothing to find
2. Compute the autocorrelation, and the naive and seasonal-naive baselines. **These are the numbers to beat**
3. Fix the protocol before fitting anything: horizon, rolling origin, number of folds, metric
4. Fit a linear model on lags. It is fast, and it is often close
5. Only then a recurrent model, scored on the same protocol
6. Report the ladder with its spread, not the best row alone

✗ Steps 2 and 3 come before step 4 for one reason: after you have a number, you will not choose the protocol that makes it look worse.

The notebook for this lecture

Open Lecture 16 in Colab

- **Runs on free CPU.** The series is short and the networks are small
- The recurrent model of today's first slides, written and trained line by line; the multivariate frame and its `shift(-1)`
- The ladder: six models, each recipe chosen inside the training period, scored on the same 151 forward days — with both selection scores printed, so you can see the near-tie that decides depth
- The simple RNN on shuffled folds; fourteen days ahead; and March 2020

> **WHAT TO CHANGE**
>
> Change the seed in the setup cell and re-run the ladder. Watch the simple and the deep RNN: if their order flips, you have reproduced the lecture's point about a ranking that rests on one near-tied choice.

What we did

1. Replaced a fixed window with a *state*: a recurrent network is a deep network whose layers share their weights
2. Saw that backpropagation through time is Lecture 11's problem with one aggravation: the same matrix multiplies the gradient at every step, so its powers vanish or explode
3. Forecast several steps ahead — recursively, directly, and sequence-to-sequence — and named exposure bias
4. Ran the same depth comparison twice and watched it flip: a ranking that rests on one near-tied choice is not a finding
5. Added more series and the next day's type, legitimately, and measured that they paid more than depth did
6. Met LSTM and GRU: a gate is a path along which the gradient is added to rather than multiplied
7. Watched an honest protocol report the regime change of March 2020 — the failure that ends forecasting systems

NOT PRESENTED — FOR AFTERWARDS

Exercises Lecture 16

five, in the style of the written paper

Exercises — Lecture 16

1 A recurrent network is trained on 56-day windows drawn from one series. Explain why shuffling the *windows* is legitimate while shuffling the *series* is not. [4]

2 A three-layer RNN beats a one-layer RNN on test and on train. A re-run of the same code reverses the order on both. State what the ranking rested on, and what would settle it. [4]

3 Gates were introduced after a plain recurrent cell failed over long windows. State the mechanism, in terms of what happens to the same matrix over 56 steps. [4]

Solutions on Lecture 17’s deck.

Exercises — Lecture 16 (continued)

4 A colleague builds the multivariate forecaster with `day_type.shift(-1)` as a feature and objects that it reads the future. State whether it is a leak, and the one question that settles it. [5]

5 Report the training MAE beside the test MAE. Name the two failures the pair distinguishes, and say why the test column alone cannot. [4]

Solutions on Lecture 17’s deck.

THE FIVE SET AT THE END OF LECTURE 15

Solutions Lecture 15

not part of this lecture

Solutions — Lecture 15

1. For a stationary series, $\mathrm{Var}(X_t - X_{t-h}) = 2\gamma(0)(1-\rho(h))$. State the condition on…

✓ $\rho(h) < 1/2$.

- The factor $2(1-\rho)$ exceeds 1 exactly then
- So the reflex 'it is not stationary, difference it' can make the problem harder

2. Explain why the forecast "copy the value from seven days ago" is hard to beat on daily ridership,…

✓ Its error is the seasonal difference $-(\nabla_7 y)_t$, with variance $2\sigma^2(1-\rho_7)$; with $\rho_7 = 0.831$ that is only $0.34\,\sigma^2$.

- The weekly cycle carries most of the structure, so ρ_7 is large
- It is not a lucky heuristic: it is the autocorrelation function read off at a single lag

Solutions — Lecture 15 (2)

3. State why MAPE is the wrong metric for transit ridership, using a specific day as the example.

✓ It divides by the truth, and a holiday's ridership is a fraction of a working day's: over the three non-working days at the end of May 2019 ridership falls to about weekend level, roughly half, so the same absolute miss counts about twice as much there.

- The metric is most sensitive exactly on the holidays, where the model is weakest
- MAE in boardings is in the units the operations team already uses

4. A cross-validated forecast uses KFold(shuffle=True) and reports a good score. Name the two conditions the…

✓ Independence: the held-out rows must be independent of the training rows. Deployment: they must come from the distribution the model will meet, where every training day precedes every forecast day.

- Neighbouring days are nearly the same number, so a shuffled test row has its own neighbours in training and is not independent of it
- Shuffling puts a point's own future in the training set, which never happens in deployment

Solutions — Lecture 15 (3)

5. A dataset class turns a pool of 1,247 daily values into windows of 56 days, each labelled with the day that…

✓ $1{,}247 - 56 = 1{,}191$ rows. One index earlier, the label is the last day of its own window: the model is handed the answer as an input, and the score measures a leak, not a forecast.

- The first 56 days have no complete window before them, so they cannot be labels
- The single index `end = idx + window_length` is both the end of the window and the position of the label; nothing else complains if it is off by one

LECTURE 17 · GÉRON CH. 14

Text

Reviews as data, and two bag-of-words baselines to beat
Softmax and cross-entropy, and why their gradient is $\mathbf{p} - \mathbf{y}$
Subword tokens, embeddings and a recurrent classifier — then the same model with pretrained embeddings swapped in

Run the Lecture 16 notebook first